

SAHAYOGI

Bridging Gaps through Skill Exchange



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Keywords



Figure 1: Keywords Word Cloud - Technical Vocabulary of Sahayogi

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1. Introduction

1.1 What is the Project?

Sahayogi is a web-based credit-based skill sharing platform designed to let people teach and learn practical skills without direct cash payment for every lesson. The name Sahayogi means helper or collaborator in Nepali, which reflects the purpose of the system: to create a trusted community where learners and teachers can exchange knowledge through a fair internal credit mechanism. In the platform, a user can post a skill they know, such as web design, Python programming, public speaking, video editing, or academic support, and other users can apply to learn that skill by spending platform credits. The same account can be used for both teaching and learning, so a person is not locked into one role. This design supports the idea of peer learning, where people learn through participation, collaboration, and shared experience rather than only through formal paid courses (Lave & Wenger, 1991).

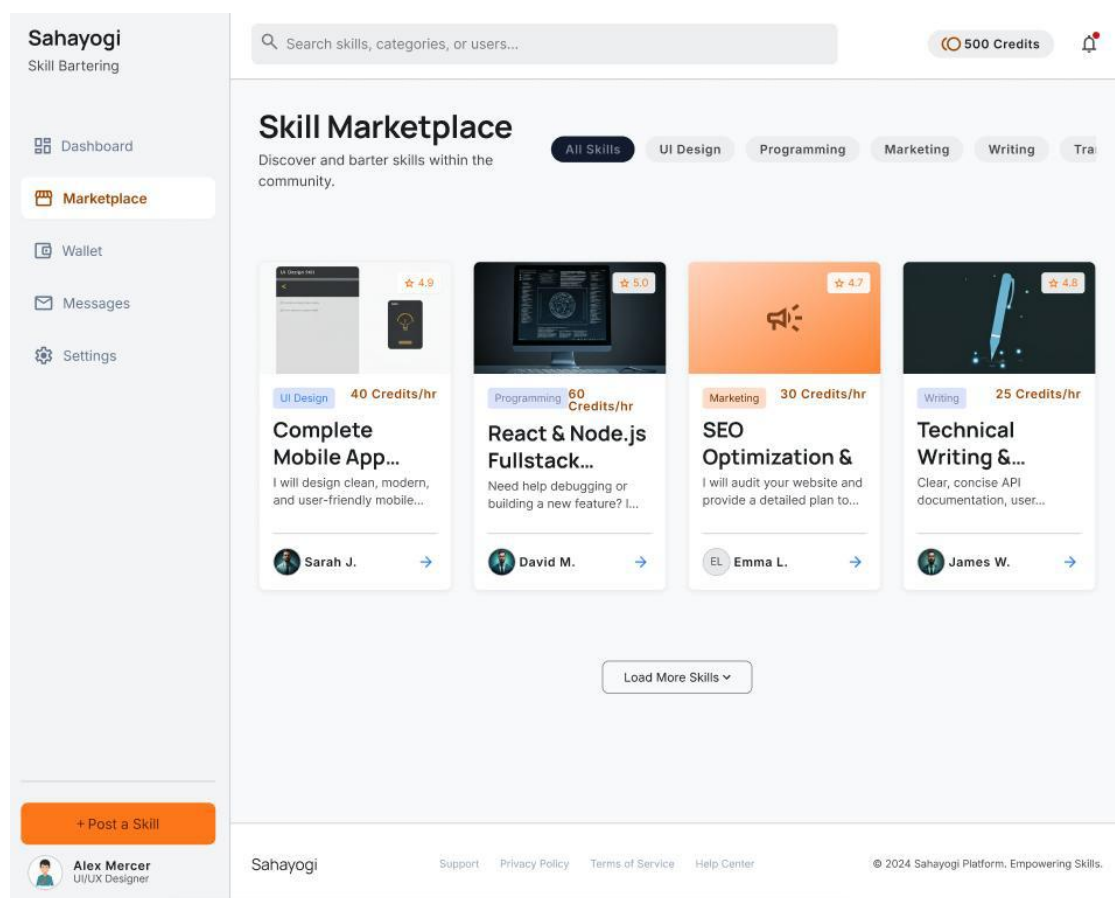


Figure 2: Introduction Interface Image - Sahayogi Skill Marketplace Homepage

1.2 Why is the Project Needed?

Many students, job seekers, and low-income learners want to build new skills but cannot always afford subscriptions, coaching fees, or premium learning platforms. At the same time, many people already possess useful skills that could help others, yet they do not have a structured way to teach those skills and receive value in return. Existing learning platforms usually depend on direct payment, while informal skill sharing between friends or classmates often lacks visibility, trust, scheduling, and accountability. Research on collaborative consumption and the sharing economy shows that people participate when a platform provides convenience, social value, and

a clear benefit for both sides (Hamari et al., 2016; Sutherland & Jarrahi, 2018). Sahayogi responds to this need by using internal credits as a non-cash exchange system. Users can earn credits by teaching and spend credits when they want to learn from others, making learning more accessible and community driven.

1.3 How Will the System Work?

The system will operate as a credit-based skill marketplace. First, a user registers, logs in, and creates a profile. The user can then post a skill they are able to teach with a category, description, availability, and credit cost. Learners browse or search for the available skills and apply for a suitable learning session. The teacher reviews the application and can accept or reject it. When the request is accepted, the users can communicate through in-app messaging or video calling. After the learning session is completed, credits are transferred from the learner to the teacher and stored in a credit transaction history. Both users can then leave reviews and ratings, helping the platform build trust and reputation over time. This workflow makes the platform different from direct barter because learners do not need to find a perfect one-to-one skill swap; they can use credits earned from one user to learn from another.

2. Aim

The aim of Sahayogi is to design and implement a secure credit-based skill sharing platform where users can teach skills to earn credit and spend credits to learn from others.

3. Objectives

- Develop secure registration, login, logout, and password protection.
- Allow users to create profiles and post teachable skills.
- Enable learners to browse, search, and apply for skills.
- Implement credit balance, transfer, and transaction history.
- Support messaging, video calling, reviews, and reputation.
- Provide admin moderation, reporting, and account deletion.

4. Problem Statement

The main problem addressed by Sahayogi is that affordable skill learning is still difficult for many learners, even when useful knowledge already exists inside their community. Paid platforms and coaching services can be expensive, while free informal learning is often unorganised and difficult to trust. Learners may not know who can teach a skill, teachers may not receive any value for their time, and both sides may lack a safe communication and review system. A credit-based platform can solve this by creating a fair internal reward system where teaching generates credits and learning consumes credits. This approach is related to community currency and time-banking ideas, where non-cash units are used to represent contribution and reciprocity (Lasker et al., 2011; Seyfang, 2002).

Problem Statement Infographic

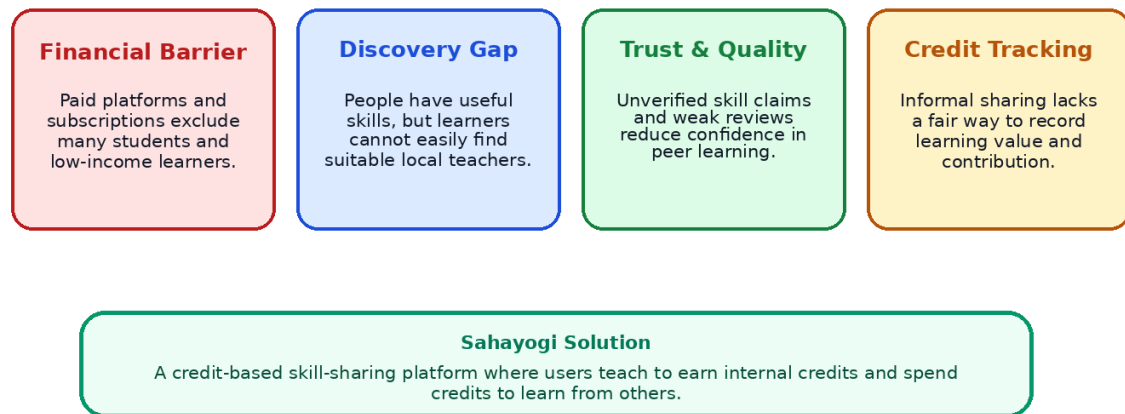


Figure 3: Problem Statement Infographic - Barriers Addressed by Sahayogi

5. Data and Research (Literature Review)

5.1 Overview of Existing Systems

Existing systems show that there is demand for online learning and digital skill exchange, but no reviewed platform fully matches the proposed Sahayogi model. Subscription learning platforms such as Skillshare and LinkedIn Learning provide structured learning content, but they are normally built around paid access. Freelance marketplaces allow people to sell skills, but their focus is service delivery for money rather than peer learning. Time-banking and community currency systems demonstrate that non-cash credit models can support reciprocal contribution, but they are often not designed as modern skill learning platforms with skill listings, teacher approval, chat, video calling, reviews, and an integrated credit ledger. Therefore, Sahayogi combines three ideas: peer learning, a digital marketplace, and an internal credit system.

Table 1: Comparison of Existing Learning and Skill-Sharing Platforms

Platform / Model	Main Strength	Limitation	Relevance to Sahayogi
Skillshare / paid learning	Large library of learning content	Subscription-based and not credit-based	Shows demand for structured digital learning
Freelance marketplaces	Connects skilled people with customers	Money-focused rather than learning-focused	Shows value of skill listings and profiles
Time banking / community currency	Supports non-cash contribution and reciprocity	Often lacks learning workflow and modern UX	Supports the internal credit concept
Sahayogi / proposed system	Users can teach, earn credits, and learn	Requires active community and moderation	Combines learning, credits, trust, and communication

5.2 Research Findings

The literature supports the key design choices of Sahayogi. Sharing economy research explains how digital platforms can coordinate access to resources, services, and skills between users (Sutherland & Jarrahi, 2018). Peer learning and communities of practice research explain why people can learn effectively from other members of a community when they communicate, participate, and exchange practical knowledge (Abedini et al., 2021; Lave & Wenger, 1991). Community currency and time-banking research supports the idea that non-cash units can represent contribution and encourage people to help one another (Lasker et al., 2011; Seyfang, 2002). Finally, research on trust and reputation systems shows that ratings, reviews, and visible feedback help users make safer decisions in online services (Jøsang et al., 2007; Liu & Munro, 2012).

5.3 Research Relevance to Sahayogi

The research directly informs the system requirements. The skill marketplace is justified by sharing economy and platform literature because the system must connect users efficiently. The credit ledger is justified by community currency research because it records the value exchanged without using money. Messaging and video calling are justified by online collaborative learning research because learners and teachers need direct interaction. Reviews, reputation scores, skill proof, and admin moderation are justified by trust-system research because users need confidence before learning from strangers online. GDPR-compliant deletion is also relevant because user platforms must treat personal data carefully and give users control over their information (European Union, 2016).

6. Features and Functionalities

6.1 Functional Requirements

The functional requirements are designed to match the research findings and the actual credit-based model of Sahayogi. Each function supports learning access, fair credit exchange, trust, communication, or safe administration.

Table 2: Functional Requirements Linked to Research Relevance

Module	Functional Requirement	Research / Design Relevance
User Access	Registration, login, logout, profile editing, and account deletion	Secure identity and data control support trust and privacy.
Skill Marketplace	Post skills, browse listings, search, and filter by category	Platform literature supports organised discovery between users.
Learning Requests	Learner applies and teacher accepts or rejects requests	Creates structured learning coordination instead of informal messaging only.
Credit System	Credit balance, credit transfer, and transaction history	Community currency research supports non-cash contribution tracking.

Communication	In-app messaging, unread badges, and video call support	Collaborative learning requires interaction between teacher and learner.
Trust	Reviews, ratings, reputation, reports, and certificate upload	Trust and reputation systems reduce uncertainty in online services.
Administration	Admin dashboard, category management, listing approval, and bans	Moderation supports safety and quality control.

6.2 Non-Functional Requirements

The non-functional requirements define the quality conditions needed for Sahayogi to be secure, usable, reliable, and maintainable. These requirements are important because a platform that handles user accounts, learning requests, credit balances, and communication must protect users and remain easy to use.

Table 3: Non-Functional Requirements

Requirement	Description
Security	Passwords will be hashed, sessions will be protected, and role-based access control will separate users and admins.
Usability	The interface will be responsive and simple so users can post, apply, and manage credits easily.
Performance	Common pages and API actions should respond quickly during normal student project use.
Scalability	The modular architecture allows future expansion of skills, users, credits, and communication features.
Reliability	Credit transactions and request statuses must be stored consistently in the database.
Privacy	Users should be able to delete accounts and minimise unnecessary personal data storage.

7. Methodology

7.1 Chosen Methodology: Agile Scrum-Inspired

An Agile Scrum-inspired methodology is suitable for Sahayogi because the project includes research, interface design, backend development, database work, testing, and revision. Scrum encourages iterative development through short cycles, review, and adaptation (Schwaber & Sutherland, 2020). This is useful for a student project because the team can build the system module by module, test early, and improve the design based on feedback.



Figure 4: Agile Scrum Methodology Used for Iterative Development

7.2 Sprint Plan

Table 4: Sprint Plan

Sprint	Main Focus	Key Deliverables
Sprint 1	Foundation	Registration, login, logout, password hashing, profiles, and roles
Sprint 2	Skill Marketplace	Skill posting, browsing, search, category filter, and listing approval
Sprint 3	Requests and Credits	Learner applications, teacher approval, credit balance, transfer, and history
Sprint 4	Communication and Trust	Messaging, video calling, reviews, ratings, reputation, and reports
Sprint 5	Admin and Finalisation	Admin dashboard, category management, GDPR deletion, testing, and polishing

7.3 Team Collaboration and Tools

The team will collaborate using GitHub for version control, shared sprint boards for planning, and scheduled review meetings for progress checks. Feature branches will be used so that incomplete work does not affect the main version. Testing checkpoints will be completed before merging important features. This process supports teamwork, reduces integration problems, and helps the team prepare a more reliable final prototype.

8. Tools and Technology

The technology stack remains the same as originally planned. It is suitable for a student web application because it is lightweight, familiar, and practical for building a database-driven platform. Flask provides the backend web framework, MySQL stores structured data, Jinja2

supports server-side templates, Bootstrap improves responsive interface design, and testing tools such as pytest help validate functionality (Bootstrap Authors, n.d.; Oracle, n.d.; Pallets, n.d.; pytest development team, n.d.).



Figure 5: Technology Stack Used for Sahayogi Development

Table 5: Tools and Technologies Used in Sahayogi Development

Category	Technologies and Tools
Frontend / GUI	HTML5, CSS3, Vanilla JavaScript, Bootstrap, Jinja2
Backend / API	Python 3, Flask
Database	MySQL
ORM / Data Access	Flask-SQLAlchemy
Database Management	MySQL Workbench
IDE / Environment	Visual Studio Code
Testing and Tooling	pytest, pip, Flask-Migrate

9. Conceptual Diagram

9.1 System Architecture

Sahayogi follows a three-tier architecture consisting of a presentation layer, an application layer, and a data layer. The presentation layer is responsible for the user interface. The application layer manages authentication, skills, learning requests, credit transfers, messaging, reviews, and admin functions. The data layer stores users, skills, requests, credits, messages, reviews, categories, and reports in MySQL.

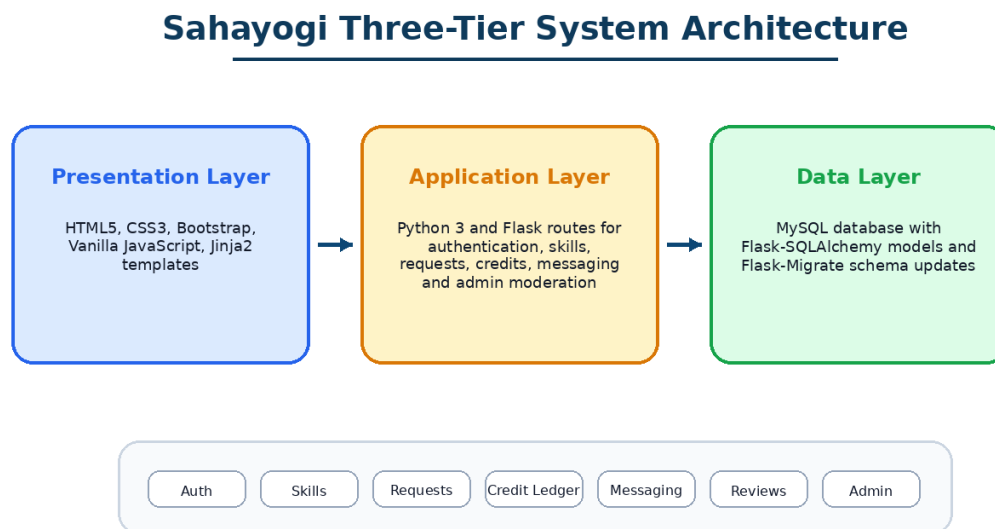


Figure 6: System Architecture Diagram - Three-Tier Architecture of Sahayogi

9.2 Entity Relationship Diagram

The main database entities include users, skills, categories, learning requests, conversations, messages, video sessions, reviews, reports, and credit transactions. A user can post many skills and can also apply to learn many skills. Accepted requests create communication records and later produce reviews and credit transactions. The credit transaction table is important because it stores every debit and credit entry, making the internal economy transparent.

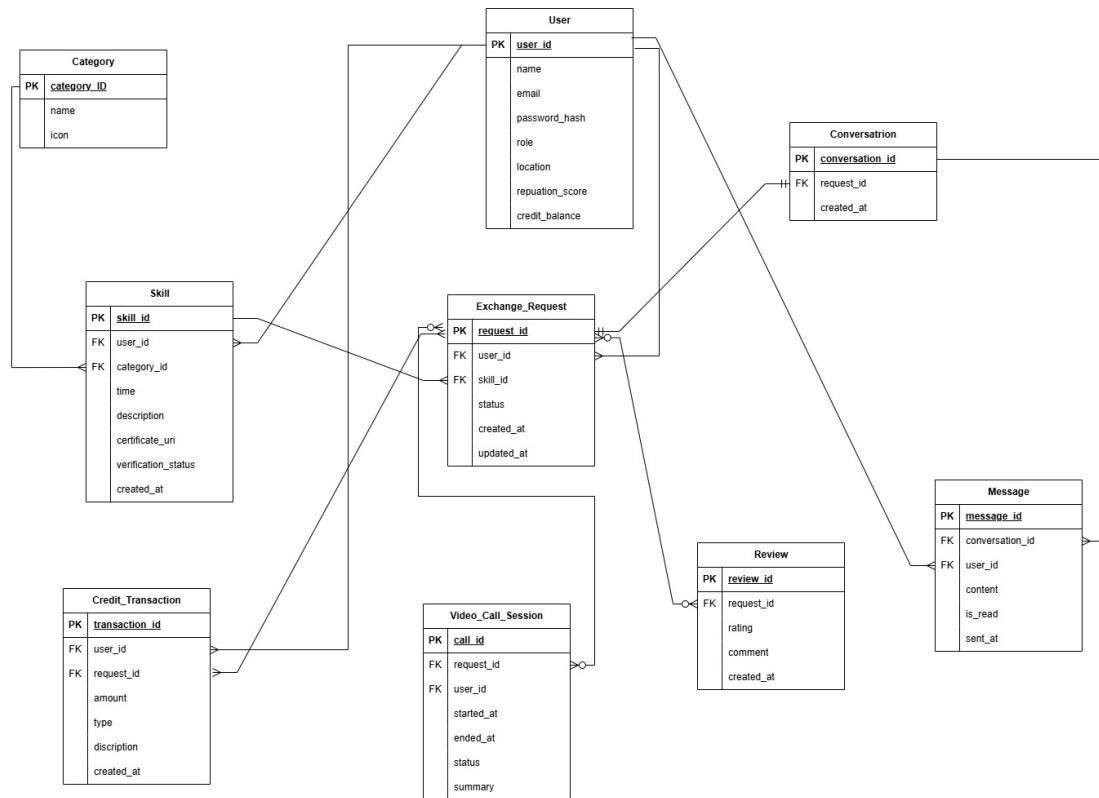
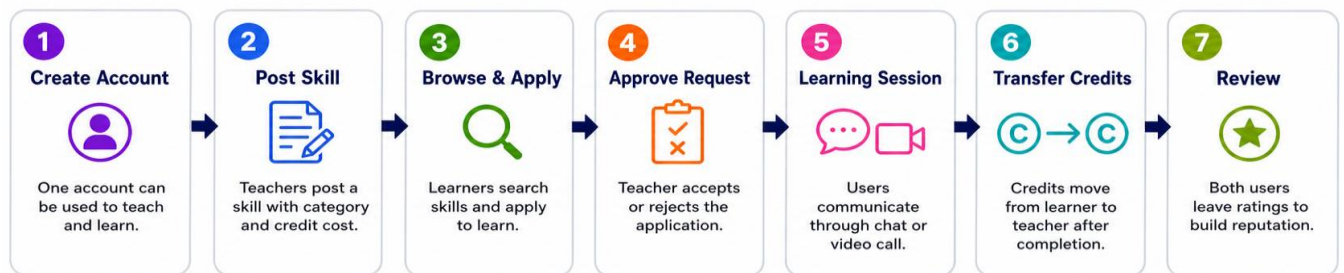


Figure 7: Entity Relationship Diagram - Data Model of Sahayogi

9.3 Credit-Based Learning Workflow

The workflow is based on a credit cycle rather than a direct mutual barter match. A teacher posts a skill and sets the credit cost. A learner applies to learn and waits for approval. After the session is completed, the system transfers credits from the learner to the teacher. The teacher can later spend those credits to learn from another user. This means contribution and learning are connected through a flexible platform credit system.

Credit-Based Skill Sharing Workflow



Platform Credit Cycle

Teach skills -> earn credits -> spend credits to learn other skills. This allows a user to be both a teacher and learner from the same account.

Figure 8: Credit-Based Learning Workflow - Teach, Earn Credits, Learn

10. Scope

10.1 Included in Scope

- Secure authentication with user, moderator, and admin roles.
- Skill listing, browsing, searching, filtering, and admin approval.
- Learner applications and teacher accept or reject workflow.
- Internal credit balance, credit transfer, and transaction ledger.
- Messaging, video calling, reviews, reputation, and reporting.
- Admin dashboard, category management, account deletion, and moderation.

10.2 Excluded from Scope

- Real money payment gateways and external financial transactions.
- Advanced AI recommendation engines or predictive matching.
- Third-party automated certificate verification APIs.
- Social media login and external social sharing features.
- Full multilingual localisation beyond the current English interface.
- Complex gamification such as badges, levels, and experience points.

11. SWOT Analysis

The SWOT analysis summarises the strengths, weaknesses, opportunities, and threats of Sahayogi as a credit-based skill sharing platform. It shows that the project has strong community-learning potential, but also requires good moderation, active users, and careful credit management.

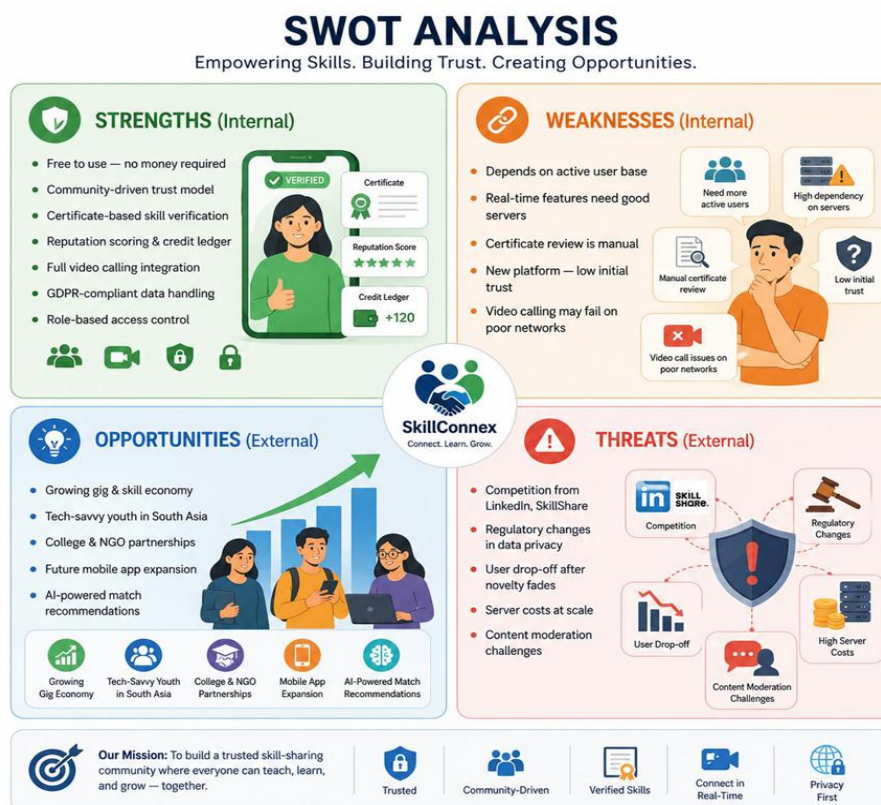


Figure 9: SWOT Analysis Matrix - Strategic Evaluation of Sahayogi

12. Conclusion

Sahayogi is a feasible and useful credit-based skill sharing platform that addresses the challenge of affordable access to practical learning. Instead of depending on direct payment or strict one-to-one barter, the platform allows users to teach skills, earn internal credits, and spend those credits to learn from others. This makes the system flexible because one account can support both teaching and learning. The proposal is supported by research on peer learning, digital sharing platforms, community currencies, and trust systems, which justifies the key features such as skill listings, learner applications, credit transactions, messaging, video calling, reviews, reputation, and admin moderation. The selected technology stack of HTML5, CSS3, Vanilla JavaScript, Bootstrap, Jinja2, Python Flask, MySQL, Flask-SQLAlchemy, MySQL Workbench, pytest, pip, and Flask-Migrate is appropriate for building a secure and manageable student web application. The Agile Scrum-inspired methodology also makes the development plan realistic because the team can complete the system through short sprints and regular testing. Overall, Sahayogi has the potential to create a fair learning community where users contribute knowledge, receive value through credits, and gain access to skills that may otherwise be difficult or expensive to learn.

13. References

- Chiu, C.-M., Hsu, M.-H., & Wang, E. T. G. (2006). Understanding knowledge sharing in virtual communities: An integration of social capital and social cognitive theories. *Decision Support Systems*, 42(3), 1872–1888. <https://doi.org/10.1016/j.dss.2006.04.001>
- Dellarocas, C. (2003). The Digitization of Word of Mouth: Promise and Challenges of Online Feedback Mechanisms. *Management Science*, 49(10), 1407–1424. <https://doi.org/10.1287/mnsc.49.10.1407.17308>
- Flask. (n.d.). *Welcome to Flask — Flask Documentation (1.1.x)*. Flask.palletsprojects.com. <https://flask.palletsprojects.com/>
- Hamari, J., Sjöklint, M., & Ukkonen, A. (2016). The Sharing Economy: Why People Participate in Collaborative Consumption. *Journal of the Association for Information Science and Technology*, 67(9), 2047–2059. <https://doi.org/10.1002/asi.23552>
- Lave, J., & Wenger, E. (1991). *Situated Learning: Legitimate Peripheral Participation*. Cambridge University Press. <https://doi.org/10.1017/cbo9780511815355>
- Lee, C., Burgess, G., Kuhn, I., Cowan, A., & Lafortune, L. (2020). Community exchange and time currencies: a systematic and in-depth thematic review of impact on public health outcomes. *Public Health*, 180, 117–128. <https://doi.org/10.1016/j.puhe.2019.11.011>
- Michel, A., & Hudon, M. (2015). Community currencies and sustainable development: A systematic review. *Ecological Economics*, 116, 160–171. <https://doi.org/10.1016/j.ecolecon.2015.04.023>
- Rajabion, L., Nazari, N., Bandarchi, M., Farashiani, A., & Haddad, S. (2019). Knowledge sharing mechanisms in virtual communities: A review of the current literature and recommendations for future research. *Human Systems Management*, 38(4), 365–384. <https://doi.org/10.3233/hsm-190516>
- Schwaber, K., & Sutherland, J. (2020). *The Scrum Guide*. <https://scrumguides.org/docs/scrumguide/v2020/2020-Scrum-Guide-US.pdf>
- Seyfang, G. (2004). Time banks: rewarding community self-help in the inner city? *Community Development Journal*, 39(1), 62–71. <https://doi.org/10.1093/cdj/39.1.62>

Appendix

<https://github.com/13hugol/sahayogi>

<https://trello.com/b/OSGHyp0k/prd>

<https://docs.google.com/spreadsheets/d/1atDIYcTWJcOQm1I8czQJFAliCsuiGWfSpseJppLpLc8/edit?usp=sharing>

<https://www.figma.com/proto/AT9KH4ac77UEBDFgMNHLpj/Untitled?node-id=0-1&t=4MIv0737P4sTgxYQ-1> (preview)

<https://www.figma.com/design/AT9KH4ac77UEBDFgMNHLpj/Untitled?node-id=0-1&m=dev&t=A8djIHcTeEo3cLOf-1> (developer)